What are forces?

A force can be thought of as a push or a pull. Every time you push or pull on something, you are exerting a force. For instance, when you ride a bike, your legs are exerting a force by pushing down on the pedals.

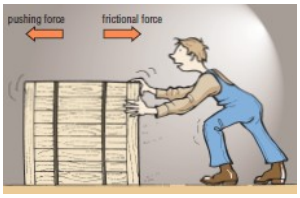
Effects of forces

Scientists say that when a force acts on something, a change occurs. Speeding up, slowing down, changing direction and changing the shape of something are examples of changes which can be produced by forces.

Contact Forces occur when one object or person exerts a force through contact with another object.

Friction

Friction is an example of a contact force. Friction occurs between any two or more surfaces that are in contact. The direction of the frictional force is normally opposite to the movement of the object. Friction is the force that opposes the movement of one body over another.



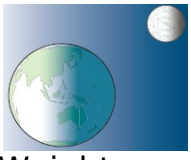
The effects of frictional forces can be an advantage or a disadvantage. Without friction, we would not be able to walk or hold objects. However, friction also causes moving parts to wear or can generate significant amounts of heat.

Some ways to reduce the effects of friction are to add a lubricant, make the surface as smooth as possible or reduce the force pushing the objects together.

Non-contact Forces are forces that act over a distance.

Gravity

Gravity is a pulling force that exists between things that have mass. The strength of the force of gravity between two objects is determined by their mass. Earth has a very large mass so it exerts a large gravitational force on objects close to it. The gravitational force of attraction increases as the mass of the objects increases or the distance between them decreases.



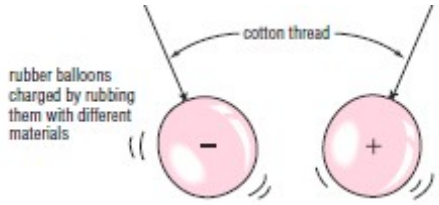
Weight can be defined as the amount by which an object is pulled downwards by gravity. It is expressed in newtons (N) and can be measured with a spring balance. Weight is different to mass. Mass is defined as the amount of matter that makes up an object and is measured in kilograms (kg).

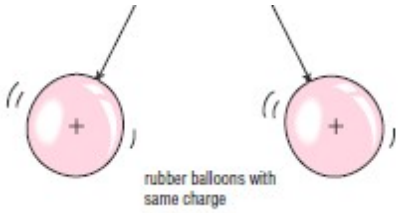
Electric

Electric forces result from the pulling or pushing effects of electrical charges. Electrical charges are the result of protons and electrons within atoms. Protons carry a positive charge and electrons carry a negative charge. When the number of electrons and protons in an atom are equal, the atom has no charge. If an atom has unequal numbers of protons and electrons, then it will have an overall charge. More

electrons than protons gives an atom a negative charge and more protons than electrons gives it a negative charge.

Objects have a positive or negative charge because the atoms on their surface have either an overall positive or negative charge. How charged objects affect each other is governed by the charge law. According to the charge law, objects that have opposite charges exert pulling forces on each other. These objects attract one another. Objects that have the same type of charge exert pushing forces on each other. These objects repel each other.





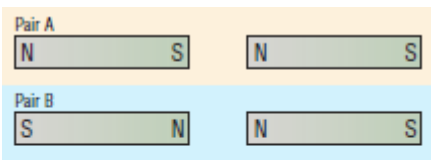
Magnetic

Magnets can affect one another or attract certain metals without being close to them. All magnets contain one or more of the metals iron, nickel or cobalt, and magnets will only affect materials containing these metals. Each end of a magnet is called a pole. One pole is called north and the other is called south. When the pole of one magnet is brought close to the pole of another magnet, the two poles either repel or attract one another. Magnetic poles follow similar rules to those for electric charges. Opposite poles attract and like poles repel.

Questions



1. What is a force?
2. What changes can be produced by a force?
3. Identify an example of a situation where a force:
    a. Changes the speed of an object.
    b. Changes the shape of an object.
    c. Changes the direction of an object.
4. In the picture of the cricketer, identify as many forces as you can. For each force, identify what is providing the force, what the force is acting on and what the effect of the force is.
5. What is a contact force?
6. What is friction?
7. Give two examples of situations where friction is an advantage.
8. Give two examples of situations where friction is a disadvantage.
9. What is a non-contact force?
10. Why is gravity considered to be a non-contact force?
11. What factors affect the strength of a gravitational force?
12. How is the mass of a pencil different from its weight?
13. Would you expect the gravitational force exerted by the moon to be smaller or larger than the gravitational force of the earth? Explain your answer.
14. How would your weight on the moon compare with your weight on earth?
15. Why are electrical and magnetic forces considered non-contact forces?
16. What is the charge law?



17. What are the metals which have magnetic properties?
18. Look at the picture on the right.
    a. Which pairs of magnets would attract each other?
    b. Which pairs of magnets would repel each other?